AI-Assisted Exam Question Analytics and Weakness-Aligned Practice Recommendation: A Repository Case Study

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Abstract- This paper presents an artifact-centered case study of an existing AI-assisted exam analytics prototype, not a new ranking algorithm or forecast of examination content. It classifies questions by revised Bloom level, derives weakness profiles from graded evidence, and recommends practice. On a small, repository-authored eleven-question integration fixture, including one prospectively specified Evaluate item, the deployed ModernBERT classified all items correctly (accuracy and six-class macro-F1 = 1.00); this is not an estimate of broader generalization. The fixture has zero exact normalized overlap with the 8,767-question training CSV. Across 100 candidates and ten synthetic weakness profiles, default P@5/NDCG@5 was 0.360/0.420 and 0.320/0.414 for two project-affiliated raters, versus random baselines of 0.320/0.349 and 0.291/0.357. A weakness-only baseline matched P@5 and had NDCG higher by 0.016 and 0.015. Paired-profile bootstrap 95% intervals for default-minus-baseline NDCG were [-0.112, 0.059] and [-0.111, 0.059], so no ranking difference was established. The case demonstrates why hand-tuned educational rankers require explicit baselines and how kappa with rank correlation can expose rater-scale miscalibration. Its contribution is reproducible system evaluation and failure analysis, not evidence of educational effectiveness.

Keywords- learning analytics, Bloom taxonomy, educational recommender systems, exam analytics, adaptive practice, repository case study

I. INTRODUCTION

Assessment evidence can do more than report a final mark. Question text, rubric criteria, awarded marks, and course context can jointly reveal which topics and cognitive processes require additional instruction. This is consistent with learning analytics as the use of learner data to support actionable teaching and learning decisions [1], [2]. Closing the loop remains difficult: a diagnostic dashboard may identify a weak topic, but lecturers and students still need suitable follow-up practice at an appropriate cognitive level.

The examined AdaptiveExamAnalyticsEngine implements this loop for a database-management course. It ingests graded submissions, attaches a Bloom level and canonical topic to each question, computes item-, student-, and cohort-level evidence, derives weakness scores, ranks a question bank, and can ask a local large language model (LLM) to generate questions or teaching actions. Numeric assessment evidence remains deterministic; probabilistic models are restricted to semantics and content generation.

This is an artifact-centered systems case study and audit, not a novel-algorithm paper. Its contributions are:

- a repository-grounded characterization of an evidence-to-action architecture and its operational failure modes;
- a six-class temporal Bloom-classifier check, plus random and weakness-only ranking baselines, paired bootstrap uncertainty, and a one-factor weight sensitivity analysis; and
- an explicit analysis of rater calibration and data imbalance that yields reusable evaluation lessons and defines the evidence required before educational deployment.

The scope is intentionally narrower than exam-question prediction. The implementation estimates learning need and the suitability of practice candidates; it contains no time-series, recurrence, or probabilistic model of future exam content. The current repository name, AdaptiveExamAnalyticsEngine, is therefore materially more accurate than the earlier prediction-oriented name.

II. RELATED WORK

The revised Bloom taxonomy organizes cognition as Remember, Understand, Apply, Analyze, Evaluate, and Create [1]. Learning analytics connects such structured evidence to instructional decisions [2]. Adaptive educational systems traditionally build learner models from goals, preferences, and knowledge, then alter navigation, content, or support [3]. Context-aware learning recommenders extend this idea by representing learner, resource, activity, and environmental context rather than relying only on generic preference signals [4].

Educational recommender research includes ontology-based and hybrid methods that combine domain knowledge, collaborative signals, and sequential access patterns [5], [6]. Personalised systems such as APELS align resources to curricula and learner needs [9], while ElectronixTutor connects knowledge-component estimates to next-resource recommendations [10]. Tag-based tutoring recommenders provide another route to personalization [11]. These systems motivate learner-aware ranking, but they also expose a distinction central to this study: a transparent heuristic may be deployable and auditable without yet being empirically superior to a simpler baseline.

Knowledge tracing estimates a learner's changing mastery from interaction sequences [7]; recurrent approaches such as Deep Knowledge Tracing learn temporal state representations from large traces [8]. The examined engine does neither. It aggregates completed assessment evidence into a current weakness snapshot and applies a fixed scoring rule. Its contribution is integration and auditability rather than a learned student-state or ranking model.

Question cognition is assigned by a ModernBERT sequence classifier [12], while optional Sentence-BERT embeddings support semantic similarity and clustering [13].

Automatic question generation has a substantial educational literature, but evaluations often under-specify pedagogical quality [14]; Bloom-targeted LLM authoring is most defensible with teachers in the validation loop [15]. Ranked outputs are evaluated with NDCG [16], and rater agreement with Cohen's kappa [17].

III. System Design

Figure 1 summarizes the implemented pipeline. Inputs are course metadata, rubrics, graded submissions, and a heterogeneous question bank. The FastAPI service persists canonical student analytics, exam analytics, generated questions, and teaching actions in MongoDB. A cache avoids repeated question classification within a rubric.

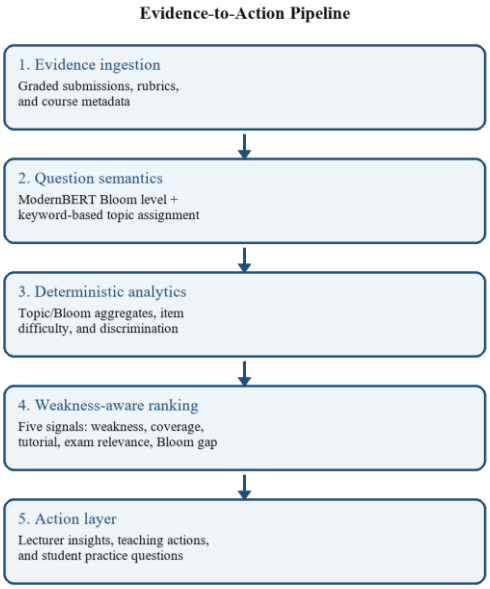


Fig. 1. Evidence-to-action pipeline implemented by the engine.

A. Question Semantics

Each question receives one of six Bloom labels (BT1-BT6) from a local ModernBERT classifier. The service exposes top-class probability as confidence and uses a 512-token default input limit. Topic assignment is deterministic: DBMS keyword counts map to eleven canonical topics, including SQL, schema refinement, security, JDBC, transactions, recovery, and indexing. Alias maps normalize noisy labels; optional MiniLM embeddings and agglomerative clustering support similarity workflows.

B. Deterministic Analytics

Awarded and maximum marks are aggregated at question, topic, Bloom, student, and cohort levels. The engine reports status bands, item difficulty, top-versus-bottom discrimination, and missed-criterion rates. Topic weakness is the complement of observed achievement. With richer evidence, the implementation uses:

$$W = 0.60(1 - p/100) + 0.25f + 0.15m \qquad (1)$$

where p is average percentage, f is failure rate, and m is missed-criterion rate. These coefficients are transparent engineering priors, not learned parameters. The repository contains no independent topic-weakness outcome against which Eq. (1) can be calibrated; the paper therefore makes no optimality claim for them.

C. Recommendation and Generation

For candidate q, the default recommendation score is:

$$R(q) = 0.35W + 0.20L + 0.15T + 0.15E + 0.15B \qquad (2)$$

where W is topic weakness, L lecture coverage, T tutorial evidence, E recent-exam relevance, and B a Bloom-level gap. These weights are also hand-set. High and Medium priorities begin at 0.65 and 0.40. The LLM may generate follow-up items after ranking; schema validation, retries, and deterministic fallbacks constrain degraded operation.

IV. Evaluation Design

A. Repository Case-Study Evidence

The question bank contains 179 records across eleven DBMS topics: 90 lecture, 55 tutorial, 18 generated, and 16 prior-exam items. It is cognitively imbalanced: 132 items (73.7%) are Apply, 39 (21.8%) Understand, and 8 (4.5%) Analyze; none are Remember, Evaluate, or Create. The rating corpus contains 100 candidates arranged as ten synthetic weakness profiles of ten candidates each. Two project-affiliated raters, lec01 and lec02, independently completed the same rating sheet. This design is a pilot self-evaluation, not independent validation.

B. Temporal Classifier Fixture

The deployed checkpoint in the working copy is dated 29 August 2026. A newer rubric fixture dated 31 August contains ten DBMS questions with repository-assigned Bloom labels: two examples each for BT1, BT2, BT3, BT4, and BT6. Before running an additional inference, we prospectively specified one BT5 Evaluate item asking the model to judge strict two-phase locking against snapshot isolation for a high-contention banking workload using serializability, deadlock risk, write skew, and throughput. Its wording was evaluated once and was not revised after observing the prediction. All eleven items have zero exact normalized text overlap with the 8,767-question training CSV. The set is treated as a small temporal repository holdout, but it was not pre-registered, is lexically explicit, and is not externally authored; accuracy, macro-F1, and the confusion matrix are therefore provisional integration evidence only.

C. Ranking Baselines and Sensitivity

Within each weakness profile, binary relevance is the rater's would-use decision. We report mean P@5 and standard binary NDCG@5. Two baselines are included: random ranking averaged over 10,000 fixed-seed permutations per profile, and weakness-only ranking using W for the candidate's canonical topic. For sensitivity, the weakness coefficient in Eq. (2) is set to 0.20, 0.35, or 0.50 while the other four weights retain their original ratio and sum to the remaining mass. To quantify uncertainty around the default-minus-weakness-only comparison, we resample the ten paired profile-level metric differences with replacement 20,000 times using a fixed seed and report percentile 95% bootstrap confidence intervals. These intervals describe pilot-sample instability, not population-level significance.

D. Rater Diagnostics

The five-point dimensions are overall quality, weakness fit, curriculum fit, difficulty fit, and clarity. We report kappa, exact agreement, agreement within one scale point, and Spearman rank correlation where useful. These diagnostics

distinguish ordinal consistency from marginal-distribution effects and systematic scale offsets.

V. Results

A. Classifier Performance

The classifier predicts all eleven temporal-fixture labels correctly. Accuracy and macro-F1 over the full six-class label space are 1.000. The ten original items have mean top-class confidence 0.836 (range 0.595-0.979); the prospectively added BT5 item is predicted as Evaluate with confidence 0.910. Table I shows the complete confusion matrix. This closes the structural class-coverage gap but demonstrates correct behavior only on explicit DBMS prompts, not generalization to implicit or ambiguous cognitive demands.

TABLE I. Temporal-fixture confusion matrix (gold rows, predicted columns)

Gold	BT1	BT2	BT3	BT4	BT5	BT6
BT1	2	0	0	0	0	0
BT2	0	2	0	0	0	0
BT3	0	0	2	0	0	0
BT4	0	0	0	2	0	0
BT5	0	0	0	0	1	0
BT6	0	0	0	0	0	2

B. Baselines and Weight Sensitivity

TABLE II. Ranking comparison over ten weakness profiles

Method	P1	N1	P2	N2
Random	0.320	0.349	0.291	0.357
Weakness only	0.360	0.436	0.320	0.429
W = 0.20	0.300	0.320	0.260	0.314
Default W = 0.35	0.360	0.420	0.320	0.414
W = 0.50	0.360	0.458	0.320	0.452

P1/N1 and P2/N2 denote P@5/NDCG@5 for lec01 and lec02. The default engine is modestly above random for both raters. Against weakness-only ranking, P@5 is identical for every profile, so its paired difference and 95% bootstrap interval are exactly 0.000. Mean default-minus-weakness-only NDCG differences are -0.016 and -0.015; their paired bootstrap intervals are [-0.112, 0.059] and [-0.111, 0.059]. Because both intervals cross zero, this sample is compatible with either a small advantage or disadvantage; it does not demonstrate incremental value from the four additional signals. Raising the weakness weight to 0.50 leaves P@5 unchanged and improves default NDCG by approximately 0.038 for each rater, while lowering it to 0.20 degrades both measures. The 0.50 result is exploratory and must not be adopted without a separate tuning and test split.

C. Rater Agreement and Coverage

Would-use prevalence is 32% and 29%. Kappa is 0.882 for would-use and for overall ratings dichotomized at four, while exact five-point overall kappa is 0.744. Weakness-fit kappa is only 0.208, yet Spearman correlation is 0.891 and 95% of pairs are within one point. The apparent contradiction comes from a systematic calibration shift: lec01 uses score 1 for 68 items, whereas lec02 uses score 2 for 57. The raters largely order weakness fit similarly but anchor the scale differently.

Difficulty-fit is more problematic. Kappa is -0.088, exact agreement is 48%, within-one agreement is 98%, and Spearman correlation is -0.029. Lec01 assigns score 4 to 94 of 100 items, while lec02 spreads ratings across 2-5. The negative kappa is therefore partly a marginal-concentration effect, but the near-zero rank correlation also shows that the construct was not shared reliably. A revised protocol should define difficulty fit as the gap between expected item difficulty and an explicitly stated target challenge band, provide anchored examples for all five scores, train raters, and adjudicate a calibration subset before independent scoring.

Coverage remains the dominant failure mode. SQL profiles achieve P@5 of 0.80 and 0.60, and JDBC 0.60 for both raters. Transaction and utility profiles achieve 0.00 for both; recovery is 0.00 and 0.20. Only three bank records cover each of indexing, transactions, and recovery. No ranking formula can retrieve useful content that is absent or incorrectly labelled.

VI. Discussion

A. Interpretation

The temporal classifier fixture closes the earlier paper's most visible reporting gap only narrowly. The prospectively worded BT5 item was classified correctly on its single unrevised run, giving nominal coverage of all six classes. The eleven-item set remains too small and too explicit to establish robust accuracy across institutions, implicit cognitive demands, adversarial wording, or within-class variation. A larger independently authored and leakage-audited test set remains mandatory.

The baseline experiment changes the interpretation of Eq. (2). The five-signal ranker beats random, but a one-signal weakness baseline is equally precise at five and has a slightly higher point estimate under NDCG. The wide paired bootstrap intervals prevent a directional performance conclusion; the defensible finding is that the additional signals have not demonstrated incremental value in ten profiles. More generally, hand-tuned educational ranking formulas should be benchmarked against their strongest single component before their complexity is credited. Future work should separate development and test profiles, tune weights only on development data, and compare against lexical, embedding, and learned-to-rank baselines.

B. Engineering Risks

Three implementation risks remain. First, classifier documentation describes a 0.60 confidence threshold and fallback, while active service paths always trust top-1 output and fail if the model is absent. The contract and code should be reconciled, and confidence calibrated. Second, keyword topic assignment produces label errors noted by the raters; semantic replacements require their own held-out evaluation. Third, the bank is dominated by Apply-level SQL material, producing structural retrieval failures regardless of ranker quality.

C. Threats to Validity

The study concerns one DBMS course. The temporal classifier set has eleven lexically explicit, repository-authored questions, only one BT5 item, and incomplete checkpoint-training provenance. The BT5 item was specified before its one inference but added after feedback on the initial study, so it repairs class support without providing independent validation. The ranking evaluation uses ten constructed

profiles, 100 candidates, and two project-affiliated raters; it measures perceived usefulness rather than student learning. Random and weakness-only baselines are informative but not competitive learned baselines. Weight sensitivity is post hoc, the same rating corpus is used for interpretation, and the bootstrap intervals are wide. No population-level inferential claim is made. These limitations define the work as a reproducible pilot case study rather than evidence of general educational effectiveness.

VII. Responsible Deployment

Recommendations should remain lecturer-facing decision support. Human review is essential for generated questions, topic labels, and teaching actions. Student identifiers and raw answer text should be minimized, access-controlled, and retained only for defined purposes. Institution-controlled endpoints reduce exposure, but database credentials must be removed from source defaults, rotated, and supplied through a secret manager. These controls align with human-centred guidance for generative AI in education [18].

VIII. Conclusion

AdaptiveExamAnalyticsEngine is a coherent AI-assisted exam analytics prototype, not an exam-forecasting model. The study reports a provisional six-class classifier check, explicit ranking baselines with paired uncertainty, weight sensitivity, broader educational-recommender positioning, and a direct explanation of weak rater agreement. The checkpoint is correct on an eleven-question temporal fixture, but the default ranker's additional signals have not demonstrated incremental value over weakness alone, and performance is constrained by bank coverage. The portable lessons are methodological: baseline hand-tuned multi-signal heuristics against simpler components, and interpret kappa alongside ordinal association and marginal distributions when diagnosing rater calibration. The next study should use an independently authored six-class Bloom set, a pre-specified difficulty rubric, balanced content, development/test separation, stronger baselines, and learning-outcome evaluation. At minimum, one external rater should independently score a preselected subset to test whether the rubric transfers beyond the project team.

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